ZARYAN SYED

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EDUCATION

Bachelor of Engineering, Computer Engineering Co-op Sept. 2024 – Apr. 2029
University of Guelph *Guelph, ON*

- **Relevant Coursework:** Operating Systems, Large-Scale Software Architecture, Data Structures, Object-Oriented Programming, Computer Organization, Embedded Systems

SOFTWARE & SYSTEMS PROJECTS

Engineering Test Data Platform | *Python, SQL, REST API, pytest, Docker, Git* Jun. 2026 – Sept. 2026

- Built a Python REST API with 9 endpoints to create, query, and delete hardware test results and report per-device and overall pass/fail statistics.
- Designed a normalized SQL schema backing all 9 endpoints, supporting lookups by test ID and device ID and failure-only queries.
- Wrote 68 pytest tests reaching 100% statement coverage (185/185 statements) and containerized the service with Docker.

Ontario Electricity Demand Forecaster | *Python, pandas, XGBoost, Power BI, Excel, Streamlit, pytest* Aug. 2026 – Sept. 2026

- Predicted Ontario's hourly electricity demand a day ahead with 3.4% average error, 23% more accurate than a simple baseline, by training an XGBoost machine learning model.
- Automated the collection and cleaning of 7 years of public grid and weather data by building a Python (pandas) pipeline.
- Pinpointed when forecasts are least accurate (August afternoons, 5.7% error vs 3.4% overall) by building a Power BI report with KPI cards, a date filter, and an hour-by-month heatmap driven by 10 DAX measures.

ESP32 Wireless Sensor Node | *C++, FreeRTOS, ESP32, I2C, Wi-Fi/HTTPS, PlatformIO* Jun. 2026 – Sept. 2026

- Eliminated sensor-sampling stalls caused by slow Wi-Fi uploads by designing dual-core FreeRTOS firmware that separates sensing from networking with a non-blocking queue, sustaining a steady 5-second sampling cycle and buffering up to 50 seconds of readings during network outages.
- Built an I2C sensor pipeline that packages readings as JSON and sends them over HTTPS to a REST API, with automatic Wi-Fi reconnection on dropped connections.
- Validated the system in an ESP32 simulator, delivering 50 of 51 readings (98%) to a live HTTP endpoint over a 4-minute run.

GPYOU – AI GPU Recommendation Platform | *JavaScript, Google Gemini API, HTML/CSS, JSON* Apr. 2025

- Built a GPU recommendation web app that converts user budget and use-case answers into structured prompts for Google's Gemini 2.0 Flash API, replacing manual spec comparison with a personalized ranked pick.
- Won Best UI at GDG HACKS 2025: Build with AI by designing a two-view interface that toggles from an intake form to a fully rendered recommendation page with matching 3D GPU models.
- Shipped a working prototype in a 48-hour hackathon on a 4-person team, integrating Blender/SketchUp 3D assets into the AI-generated results.

EXPERIENCE

Team Member Nov. 2022 – Present
Tim Hortons *Milton, ON*

- Coordinated tasks across teams of 10–12 employees during high-volume shifts, resolving operational issues to keep service and quality standards on pace.
- Rotated through multiple workstations covering order preparation, inventory, sanitation, and customer service, following standardized procedures on every shift.

Construction Assistant Apr. 2025 – Sept. 2025
ASW Works *Milton, ON*

- Supported crews of up to 4 across 3–5 residential renovations, adapting installations to changing site requirements to keep projects on schedule.
- Executed precision measurement, framing, drywall, and flooring using 10+ hand and power tools, delivering accurate installations.

TECHNICAL SKILLS

Programming: Python, C, C++, Java, JavaScript, MATLAB

Backend, Data & ML: REST APIs, SQL, Data Analysis, pandas, NumPy, scikit-learn, XGBoost, Time-Series Forecasting, pytest

Systems & Embedded: Linux, POSIX Threads, CPU Scheduling, Concurrency, FreeRTOS, I2C

Tools: Git, Docker, GCC, Make, VS Code, PlatformIO, Jupyter, Streamlit, Excel, Power BI (DAX, Power Query)